

PCA (Principal Component Analysis)

PCA is a dimensionality reduction technique used in Machine learning to reduce the number of features while keeping most of the important feature information (variance).

Why we use PCA?

⇒ For a dataset like it having 100 feature (column).

problem may arise:

- 1) Training becomes slow
- 2) huge memory is reqd
- 3) Some feature may contain duplicate information
- 4) Difficult is visualizing

PCA help by:

- 1) Reducing features
- 2) removing redundancy
- 3) Speeding up ML algorithms
- 4) reducing noise
- 5) making visualization easier (2D or 3D)

Important Terms

- 1) Features
- 2) variance
- 3) Principal component
- 4) Eigenvector
- 5) Eigenvalue

Eigenvector Vs Eigenvalue

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Eigenvector is direction

Eigenvalue is importance of that direction

eg: If you throw ~~stone~~ many stone on ground and they form a long oval shape,

The

* * *

* → Stone

*

*

*

*

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The longest direction of the oval is where data varies the most,

that direction → Eigenvector

how much data spread along that direction
→ Eigenvalue

Steps of PCA

- ① collect data
- ② Standardize Feature (mean=0 and std=1)
- ③ Find covariance matrix
- ④ Calculate Eigenvector and Eigenvalue
- ⑤ choose top principal component
- ⑥ Transform data into those components

Numerically

We are given 10 observation of two variable x and y .

X	2.5	0.5	2.2	1.9	3.1	2.3	2.0	1.0	1.5	1.1
Y	2.4	0.7	2.9	2.2	3.0	2.7	1.6	1.1	1.6	0.9

Step (1) : Find mean of each variable (x and y)

$$\bar{x} = \frac{\text{Summation of all for } x}{n \text{ of data}} = \frac{2.5 + 0.5 + 2.2 + \dots + 1.1}{10} = 1.81$$

$$\bar{y} = \frac{\text{Summation of all for } y}{n \text{ of data}} = \frac{2.4 + 0.7 + 2.9 + \dots + 0.9}{10} = 1.91$$

Step (2) : Deviation of mean

X	Y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(y - \bar{y})^2$	$(x - \bar{x}) \times (y - \bar{y})$
2.5	2.4	0.69	0.49	0.4761	0.2401	0.3381
0.5	0.7	-1.31	-1.21	1.7161	1.4641	1.5851
2.2	2.9	1	0.99	0.1521	0.9801	0.3861
1.9	2.2					
3.1	3.0					
2.3	2.7					
2.0	1.6					
1.0	1.1					
1.5	1.6					
1.1	0.9	-0.71	-1.01	0.5041	1.0201	0.7171
SUM =				5.549	6.449	5.839

Step ③: The Covariance matrix

$$\text{COV}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

using $n-1 = 9$ and column sums of the table:

$$\text{COV}(x, x) = \frac{5.549}{9} = 0.6166$$

$$\text{COV}(y, y) = \frac{6.449}{9} = 0.7166$$

$$C = \begin{bmatrix} \text{COV}(x, x) & \text{COV}(x, y) \\ \text{COV}(y, x) & \text{COV}(y, y) \end{bmatrix} = \begin{bmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{bmatrix}$$

Step ④: Eigenvalue finding

Formula: $C - \lambda I = 0$

$$\begin{vmatrix} 0.6166 - \lambda & 0.6154 \\ 0.6154 & 0.7166 - \lambda \end{vmatrix} = 0$$

Expand the determinant:

$$(0.6166 - \lambda)(0.7166 - \lambda) - (0.6154)(0.6154) = 0$$

$$\lambda^2 - 1.3331\lambda + 0.0620 = 0$$

Solving equation:

$$\lambda_1 = 1.2840 \quad \lambda_2 = 0.0490$$

Step 5: Eigenvector for λ_2

Repeat the same process for the second eigenvalue

$$\lambda_2 = 0.0490 \text{ with } v = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$0.6166x_2 + 0.6154y_2 = 0.0490x_2 \Rightarrow 0.6154y_2 = -0.5676x_2$$

$$x_2 = -1.0844y_2$$

Take,

$$y_2 = 1 \text{ then } x_2 = -1.0844 \text{ giving the direction } \begin{bmatrix} -1.0844 \\ 1 \end{bmatrix}$$

Normalise (Divide by its length):

$$\sqrt{1.0844^2 + 1^2} = \sqrt{1.1759 + 1} = \sqrt{2.1759} = 1.4751$$

$$v_2 = \begin{bmatrix} -1.0844 / 1.4751 \\ 1 / 1.4751 \end{bmatrix} = \begin{bmatrix} -0.7352 \\ 0.6779 \end{bmatrix}$$

Step 6: choose the principal component
Eigenvalue Eigenvector

$$v_1, \lambda_1 = 1.2840 \begin{bmatrix} 0.6779 \\ 0.7352 \end{bmatrix}$$

$$v_2, \lambda_2 = 0.0490 \begin{bmatrix} -0.7352 \\ 0.6779 \end{bmatrix}$$

Since $\lambda_1 > \lambda_2$, v_1 is the direction of maximum variance. This is the first PC kept when reducing one dimension.

Step 7: Recast Data on PC

project mean centered data onto both eigenvectors

$$PC_1 = (x - \bar{x}, y - \bar{y}) \cdot v_1$$

$$PC_2 = (x - \bar{x}, y - \bar{y}) \cdot v_2$$

$x - \bar{x}$	$y - \bar{y}$	PC_1	PC_2
0.69	0.49	0.828	-0.175
-1.31	-1.21	-1.778	0.143
0.39	0.99	0.992	0.384
0.09	0.29	0.274	0.130
0.29	1.09	1.676	-0.209
0.49	0.79	0.913	0.175
0.19	-0.31	-0.099	0.350
-0.81	-0.81	-1.145	0.042
-0.31	-0.31	-0.438	0.018
-0.71	-1.01	-1.224	0.163